

1 Topological Spaces

Definition 1.0.1. Let X be any set. A topology T on X is a collection of subsets of X , which satisfies the following axioms.

- $\emptyset$ and X are both in T
- Closed under arbitrary unions
- Closed under finite intersections

The sets in T are called open sets.

A set X equipped w/ a topology is called a topological space.

Example 1.0.2.

- X is any set, $T = P(X)$. (discrete topology)
- X is any set, $T = \{\emptyset, X\}$. (indiscrete topology)
- $X = \{a, b, c\}$, $T = \{\emptyset, X, \{a, b\}\}$
- $X = \mathbb{R}$, $T = \{\emptyset, X\} \cup \{U \subset \mathbb{R} \mid \text{for all } x \in U, \exists \varepsilon > 0 \text{ s.t. } (x - \varepsilon, x + \varepsilon) \subset U\}$

Proposition 1.0.3. *This is a topology.*

Proof.

- $\emptyset, \mathbb{R}$ are both open
- Suppose $\{U_j\}_{j \in J}$ is a collection of open sets in $\mathbb{R}$
Let $x \in \bigcup_{j \in J} U_j$. Then $x \in U_j$ for some $j \in J$.
 U_j is open thus there is an open interval in U_j containing x . So it is in the arbitrary union
- Let U, V be open subsets of $\mathbb{R}$
Let $x \in U \cap V$. Because U, V open there are intervals $(a, b) \subset U$, $(c, d) \subset V$ s.t. $x \in (a, b) \cap (c, d)$
 $(a, b) \cap (c, d) = (\max(a, c), \min(b, d)) \subset U \cap V$. Thus, x is in this interval thus $U \cap V$ is open.

□

- X is any set, let T be the collection of subsets $U \subseteq X$ s.t. either $U = \emptyset$, or $X \setminus U$ is finite.

Proposition 1.0.4. *This is a topology. (cofinite topology)*

Proof.

- The empty set is open, and $X \setminus X = \emptyset$ is finite, so $X \in T$
- Consider sets $\{U_j\}_{j \in J}$ s.t. $U_j \in T$
 First: consider the case for some j' , $X \setminus U_{j'}$ is finite.
 $X \setminus \bigcup_{j \in J} U_j = \bigcap_{j \in J} X \setminus U_j \subset X \setminus U_{j'}$. So the arbitrary union is finite.
 Second: If the first case does not occur, then $U_j = \emptyset$ for all $j \in J$, so the union is the empty set.
- Let $U, V \in T$. If either is the empty set then $U \cap V = \emptyset \in T$, so assume that both are not empty. Then consider $X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$, so it is a union of two finite sets which is finite thus $U \cap V \in T$

□

Definition 1.0.5. If X is a set with topology T , then the complement of an open set in X is called a closed set.

Example 1.0.6. If T is cofinite topology, a set in $A \subseteq X$ is closed $\iff$ either finite or $A = X$

IMPORTANT: Some sets are neither open nor closed. Some sets are both closed and open.

1.1 Basis for a topology

Definition 1.1.1. Let X be a set. A collection β of subsets of X is a basis for a topology if:

- every $x \in X$ is in some $B \in \beta$
- If $x \in B_1 \cap B_2$ for $B_1, B_2 \in \beta$
then: $\exists B_3 \in \beta$ s.t. $x \in B_3 \subseteq B_1 \cap B_2$

The topology generated by β is the collection T of subsets defined by:

$$U \in T \iff \text{for every } x \in U, \exists B \in \beta \text{ s.t. } x \in B \subseteq U$$

Example 1.1.2.

$X = \mathbb{R}$, β = set of open intervals in $\mathbb{R}$

$X = \mathbb{R}^n$, β = set of open balls $B(x, r) = \{y \in \mathbb{R}^n \mid d(x, y) < r\}$

Check that the second example is actually a basis for a topology

- every $x \in \mathbb{R}^n$ lies in some open ball. $x \in B(x, 1)$
- Assume $x \in B_1 \cap B_2 = B(y_1, r_1) \cap B(y_2, r_2)$
Set $r_3 = \min(r_1 - d(x, y_1), r_2 - d(x, y_2))$
Consider the ball $B_3 = B(x, r_3)$. Let $z \in B_3$.
 $d(z, y_1) \leq d(x, y_1) + d(z, x) < d(x, y_1) + r_3 \leq d(x, y_1) + r_1 - d(x, y_1) = r_1$
So $z \in B(y_1, r_1) = B_1$ do the same proof for B_2 thus $z \in B_1 \cap B_2$ thus $B_3 \subseteq B_1 \cap B_2$

Proposition 1.1.3. *This is actually a topology on X .*

Proof.

- $\emptyset \in T$ vacuously, and $X \in T$ comes from first basis axiom.
- Suppose $\{u_j\}_{j \in J}$ of sets in T .
Let $x \in \bigcup_{j \in J} u_j$. Thus $x \in u_{j'}$ for some $j' \in J$, so $\exists B_{j'} \in \beta$ s.t. $x \in B_{j'} \subseteq u_{j'}$ thus $x \in B_{j'} \subseteq \bigcup_{j \in J} u_j$. This shows $\bigcup_{j \in J} u_j \in T$
- Suppose $x \in u_1 \cap u_2$ for $u_1, u_2 \in T$. Then $x \in B_1, B_2$ for sets $B_1, B_2 \in \beta$ s.t. $B_1 \subseteq u_1$ and $B_2 \subseteq u_2$ thus $x \in B_1 \cap B_2$. By the second axiom thus implies $\exists B_3 \in \beta$. Thus $x \in B_3 \subseteq B_1 \cap B_2 \subseteq u_1 \cap u_2 \in T$

□

Remark 1.1.4. Every topology is a basis for itself.

Proposition 1.1.5. *Let X be a set, let β be a basis for topology T . The sets in T are precisely the sets of the form*

$$\bigcup_{j \in J} B_j$$

for $\{B_j\}_{j \in J}$ collection of sets in β

Proof.

- Every set of this form is in T :

$$x \in \bigcup_{j \in J} B_j, \text{ then } x \in B_{j'} \subset \bigcup_{j \in J} B_j$$

- Every set in T has the form:

$$\text{Let } U \in T. \text{ Then for every } x \in U, \text{ there is a } B_x \in \beta \text{ s.t. } B_x \subset U \implies U = \bigcup_{x \in U} B_x$$

□

Decomposition is NOT in general unique.

Basis sets are always open.

1.1.1 Comparing Topologies

Definition 1.1.6. If T_1, T_2 are topologies on the same set X , we say that T_1 is finer than T_2 if $T_1 \supseteq T_2$. Also, T_2 is coarser than T_1

Proposition 1.1.7. *Let β, β' be bases for topologies T, T' on X . Then $T \subseteq T'$ (T' is finer than T) if and only if every $x \in X$, and every $B \in \beta$ containing x , $\exists$ some $B' \in \beta'$ s.t. $x \in B' \subset B$*

Example 1.1.8. $X = \mathbb{R}^n$, let β be the set of open boxes in $\mathbb{R}^n$: sets of the form $(a_1, b_1) \times (a_2, b_2) \times \dots \times (a_n, b_n)$

This is a basis. It generates the same standard topology on $\mathbb{R}^n$.

Example 1.1.9. Let $X = \mathbb{R}$. Consider collections of half-open intervals in $\mathbb{R}$: $[a, b), (a, b]$.

Claim: this is a basis for a topology.

This is not the same as the std. topology on $\mathbb{R}$

This is finer than the std. topology.

1.1.2 Examples of topological spaces

Example 1.1.10 (Order Topology).

Let X be a set with the order relation $<$. For $a < b$ in X , define intervals:

$$(a, b) = \{x \in X \mid a < x < b\}$$

$$[a, b) = \{x \in X \mid a \leq x < b\}$$

Similar for $(a, b], [a, b]$

Let β be the set of all intervals in X of the forms:

- (a, b) for $a < b$ in X .
- $[a_0, b)$ where a_0 is the smallest element of X .(if it exists).
- $(a, b_0]$ where b_0 is the largest element of X .(if it exists).

Proposition 1.1.11. *This is a basis for a topology on X .*

This is called the order topology on X .

Example 1.1.12. Standard topology on $\mathbb{R}$ is order topology, on $\mathbb{R}$

Example 1.1.13 (Product Topology). Let X, Y be topological spaces.

Let β be a collection of sets:

$$\{U \times V \mid U \text{ open in } X, V \text{ open in } Y\}$$

Proposition 1.1.14. *This is a basis for a topology.*

This is called the product topology on $X \times Y$

Example 1.1.15. $\mathbb{R} \times \mathbb{R}$ with product topology is std topology on $\mathbb{R}^2$

If β_1 is a basis for topology on X , and β_2 is a basis for topology on Y , then

$$\{B \times C \mid B \in \beta_1, C \in \beta_2\}$$

is a basis for the product topology.

This definition is valid for products of pairs of sets.

Lemma 1.1.16. Suppose X is a topological space. Let C be a collection of open subsets satisfying:

- for each open set $U \subseteq X$ and each $x \in U$, $\exists c \in C$ s.t. $x \in c \subseteq U$

Then, C is a basis for top. on X .

Proof. Check C is a basis.

- Every $x \in X$ in some $c \in C$. X is open, so there exists c s.t. $x \in c \subseteq X$
- If $c_1, c_2 \in C$, then for any $x \in c_1 \cap c_2$, $\exists c_3 \in C$ s.t. $x \in c_3 \subseteq c_1 \cap c_2$.

Follows from the fact that c_1, c_2 are open which implies $c_1 \cap c_2$ is open so can find c_3

Check: C generates same topology T on X .

First, suppose U is open in T . By assumption for all $x \in U$, there exists $c \in C$ s.t. $x \in c \subseteq U$, so U is open in the topology generated by C .

If U is open in topology generated by C . Then U is a union of sets in C . But every $c \in C$ is open in T , so union U is also open in T . $\square$

Example 1.1.17 (Metric Topology).

Definition 1.1.18. Let X be a set. A function $d : X \times X \rightarrow \mathbb{R}$ is a metric if it satisfies:

- $d(x, y) = 0 \iff x = y$
- $d(x, y) = d(y, x) \quad \forall x, y \in X$
- $d(x, y) + d(y, z) \geq d(x, z) \quad \forall x, y, z \in X$

A set X equipped with a metric d is a metric space.

Let β be the set of all metric balls:

$$B_d(x, r) = \{y \in X \mid d(x, y) < r\} \quad \text{for } x \in X, r > 0$$

Proposition 1.1.19. This is a basis for a topology

This is called metric topology

Definition 1.1.20. A topological space X is called metrizable if $\exists$ metric d on X inducing topology on X .

Remark 1.1.21. There could be more than one metric inducing the same topology.